# Structured Row-Token Decoding for Line-Anchor-Free 2D Lane Detection

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***Abstract*—Lane detection is a fundamental perception task for autonomous driving. Existing methods often represent lane geometry using dense segmentation maps, hand-designed line anchors, parametric curves, or holistic transformer queries that compress the complete lane shape into a single vector. However, lanes in perspective images are long, thin, and naturally organized as ordered row-wise structures. Compressing the full geometry of a lane into a single global representation can therefore be suboptimal, especially under curvature, occlusion, shadows, low illumination, and crowded traffic.**

**We propose a structured row-token decoder for 2D lane detection. Instead of representing each lane candidate with a single holistic query, we factorize each lane into a global instance token and an ordered sequence of row-level geometry tokens. The instance token represents lane identity and lane-level decisions, while the row tokens model the lane geometry at predefined horizontal rows. Each row token reads row-local visual evidence from image features and predicts a discrete distribution over horizontal positions. The row-wise distributions are supervised with soft-bin distributional targets to improve localization precision. The resulting representation provides an explicit inductive bias for lane geometry without relying on hand-designed line anchors.**

**Preliminary experiments on CULane with a ResNet-34 backbone show that the proposed structured representation substantially improves over an unstructured query baseline and reaches a competitive performance regime without EMA or test-time augmentation. The main contribution of this work is not merely high-resolution training or distributional regression; rather, it is the explicit decomposition of lane representation into instance-level identity and ordered row-level geometry inside the decoder.**

***Index Terms*—Lane detection, autonomous driving, transformer decoder, row-wise localization, structured representation, instance token, row token, distributional regression.**

## I. Introduction

Lane detection is a key component of autonomous driving perception systems. It provides geometric information for lane keeping, path planning, map alignment, and downstream decision making. Although the task appears simple, robust lane detection is difficult in real traffic scenes. Lane markings may be partially occluded by vehicles, degraded by shadows or illumination changes, confused with road boundaries, or completely absent in intersections and no-line regions.

Many lane detection methods formulate the problem as semantic segmentation, anchor regression, row-wise classification, or parametric curve fitting. Segmentation-based methods preserve spatial detail but require additional instance separation and post-processing. Anchor-based methods can be accurate but depend on manually designed line anchors and refinement heuristics. Parametric curve methods provide compact representations but may struggle with local irregularities. Transformer-based methods reduce hand-designed priors by using learned queries, but many such methods still represent a complete lane with one holistic query vector.

This work starts from a simple observation: in a front-view image, a lane can be naturally represented as an ordered sequence of horizontal row positions. At each predefined row, the main geometric unknown is the horizontal coordinate of the lane. Therefore, a lane representation should not necessarily compress the entire geometry into one vector. Instead, it can explicitly separate global lane identity from row-wise geometry.

We propose a structured row-token decoder, denoted as SRTD. Each lane candidate is represented by an instance token and an ordered sequence of row tokens. The instance token carries lane-level identity and objectness information, while each row token specializes in predicting the lane position at a particular horizontal row. The row tokens interact with image features through row-local attention and exchange vertical context along the same lane. This produces a structured latent representation aligned with the geometry of the task.

The contributions of this work are:

- We introduce an instance-to-row token factorization for 2D lane detection, where each lane is represented by a global instance token and ordered row-level geometry tokens.
- We design a structured decoder that combines row-local visual evidence, same-row lane interaction, and same-lane vertical context to model lane geometry.
- We use row-wise distributional localization, where each row token predicts a discrete horizontal position distribution supervised by soft-bin targets.
- We define an experimental protocol to isolate the effect of the structured representation from resolution, capacity, decoder depth, and distributional supervision.

## II. Related Work

### A. Segmentation-Based Lane Detection

Early deep lane detection systems often formulate lane detection as semantic or instance segmentation. These methods use dense pixel-level supervision and can preserve fine spatial boundaries. However, they typically require additional clustering or post-processing to recover lane instances, and their computational cost can be high for real-time applications.

### B. Row-Wise Lane Localization

Row-wise methods exploit the fact that lanes in front-view images can be described by their horizontal positions at predefined vertical rows. UFLD-style approaches reformulate lane detection as a row-wise classification problem. This representation is efficient and well aligned with common lane annotations. However, purely row-wise classifiers may have limited instance-level interaction and global geometric reasoning.

### C. Anchor-Based Lane Detection

Anchor-based methods define a set of line anchors and train the model to classify and refine them. Such methods, including strong refinement-based systems, can achieve high accuracy. Their limitation is that the representation depends on anchor design, refinement stages, and post-processing choices. Our method avoids hand-designed line anchors and instead uses learnable instance tokens.

### D. Transformer and Query-Based Lane Detection

Transformer-based methods introduce learned queries to represent lane candidates. A query can attend to image features and produce a lane prediction. While this reduces the need for dense anchors, a single query may still be an overly compressed representation for a long and curved lane. Our method keeps the query-based formulation but decomposes each lane query into a global instance token and ordered row-level geometry tokens.

### E. Hierarchical and Point-Level Queries

Hierarchical query representations have appeared in related perception tasks such as vectorized map construction and 3D lane modeling. These works motivate the idea that an object can be represented by both instance-level and point-level components. Our work adapts this principle to 2D front-view lane detection by binding point-level tokens to fixed horizontal image rows, thereby removing permutation ambiguity and introducing a task-specific geometric prior.

### F. Positioning of This Work

The proposed method is closest in spirit to the intersection of row-wise lane detection and query-based transformer decoding. However, its representation is different from both a standard row-wise classifier and a standard lane-query transformer. A standard row-wise classifier predicts horizontal locations at predefined rows but does not maintain an explicit lane-conditioned token state for each row. A standard lane-query transformer maintains a lane-level query, but the entire lane geometry is compressed into that single query. In contrast, the proposed decoder maintains a structured latent state for each pair of lane candidate and image row.

Table I summarizes the conceptual difference.

## III. Method

### A. Problem Formulation

Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, the goal is to predict a set of lane instances. Each lane is represented by a sequence of horizontal coordinates at predefined vertical rows:

$$L_i = \{x_{i,r}\}_{r=1}^{R}, \tag{1}$$

where $i$ indexes the lane instance and $r$ indexes the row. Some rows may be invalid if the lane is not visible at that vertical position. The model also predicts lane-level existence, valid vertical range, and confidence information.

### B. Overview

The proposed architecture is shown conceptually in Fig. 1. A CNN backbone extracts visual features, which are fused by a feature pyramid. The structured decoder then uses instance tokens and ordered row tokens to produce lane predictions.

### C. Terminology for a CNN Reader

The terms token and query are often unfamiliar to readers trained primarily on CNN-based vision models. In this work, a token can be understood as a learnable feature vector with a role. It is not a piece of source code and it is not a cropped image patch. It is a latent vector that the network updates using visual evidence and other tokens.

Table II defines the main terms used throughout the paper.

### D. Instance Tokens

An instance token represents a lane candidate at the global level. It is responsible for maintaining lane identity and supporting lane-level decisions such as whether the candidate corresponds to a real lane. If the model uses $N$ lane candidates, it maintains $N$ learnable instance tokens:

$$\mathbf{e}_i \in \mathbb{R}^{C}, \quad i = 1, \ldots, N. \tag{2}$$

The instance token should be interpreted as a learned slot for a potential lane. It does not correspond to a fixed semantic class such as “left lane” or “right lane” before training. During training, the assignment process encourages different slots to specialize into useful lane candidates.

### E. Ordered Row Tokens

A row token represents a predefined vertical row in the image. If the lane is sampled at $R$ rows, the model maintains an ordered sequence of row tokens:

$$\mathbf{r}_j \in \mathbb{R}^{C}, \quad j = 1, \ldots, R. \tag{3}$$

Unlike generic point tokens, row tokens have a fixed order and a fixed geometric meaning. The first token corresponds to an upper row, while later tokens correspond to lower rows. This order is essential because lane geometry is continuous along the vertical direction.

TABLE I
CONCEPTUAL COMPARISON OF LANE REPRESENTATION PARADIGMS.

| Paradigm | Lane Representation | Main Strength | Limitation Addressed by This Work |
|---|---|---|---|
| Segmentation-based | Pixel-level lane mask | Preserves spatial detail | Requires instance separation and post-processing |
| Anchor-based | Predefined line anchors refined by the model | Strong accuracy with mature refinement pipelines | Depends on manually designed line anchors |
| Row-wise classification | Horizontal position per predefined row | Efficient and aligned with lane annotations | Often lacks explicit lane-conditioned row states |
| Parametric curve | Polynomial, Bezier, or spline parameters | Compact and smooth representation | Can be restrictive for irregular local geometry |
| Holistic query transformer | One query vector per lane candidate | Learnable and end-to-end | Compresses complete lane geometry into one vector |
| **Proposed** | **Instance token + ordered row tokens** | **Separates lane identity from row-wise geometry** | **Keeps a structured decoder state for the lane curve** |

**Input Image → ResNet/FPN Features → Structured Row-Token Decoder → Lane Outputs**

Image features    Instance + row tokens    Existence, range, quality, row-wise x distributions

Fig. 1. Conceptual overview of the proposed structured row-token decoder. The figure is a placeholder and should be replaced with a graphical diagram in the final paper.

TABLE II
CONCEPTUAL MEANING OF THE MAIN TOKEN TYPES.

| Term | Meaning in this paper |
|---|---|
| Instance token | A global lane-candidate vector. It represents which lane proposal is being described and supports lane-level decisions such as existence and quality. |
| Row token | A row-position vector. It represents a predefined horizontal image row and provides geometric order. |
| Lane-row token | A lane-specific row representation. It describes one lane candidate at one vertical row. |
| Row-local evidence | Visual features from the corresponding horizontal row of the image feature map. |
| Distributional x prediction | A probability distribution over horizontal bins instead of a single scalar x value. |

| | Row 1 | Row 2 | $\cdots$ | Row $R$ |
|---|---|---|---|---|
| Instance 1 | • | • | $\cdots$ | • |
| Instance 2 | • | • | $\cdots$ | • |
| $\vdots$ | $\vdots$ | $\vdots$ | $\ddots$ | $\vdots$ |
| Instance $N$ | • | • | $\cdots$ | • |

Fig. 2. Instance-to-row token factorization. Each dot is a lane-row token representing one lane candidate at one vertical row.

### F. Instance-to-Row Factorization

For each lane candidate and each row, we form a lane-row representation by combining lane identity and row position:

$$\mathbf{z}_{i,j}^{0} = \phi(\mathbf{e}_i, \mathbf{r}_j), \tag{4}$$

where $\phi(\cdot)$ denotes a lightweight fusion operation. Conceptually, this representation means:

"the geometry token of lane candidate $i$ at row $j$."

This factorization is the central idea of the method. The model does not use a single vector to store the whole lane. Instead, it creates a structured grid of latent tokens:

$$\mathbf{Z}^{0} \in \mathbb{R}^{N \times R \times C}. \tag{5}$$

Fig. 2 illustrates this representation.

### G. Why Combine Instance and Row Information?

A row token alone only knows the vertical position. An instance token alone only knows the lane candidate identity. Lane detection requires both:

$$\text{lane-row state} = \text{which lane candidate} + \text{which row}. \tag{6}$$

This is analogous to combining content and position information in transformer models. The row component provides geometric order, while the instance component makes the row representation lane-specific. Therefore, the representation can distinguish the same row for different lane candidates.

For a CNN analogy, a lane-row token can be viewed as a feature cell whose spatial coordinate is not a 2D image pixel, but a structured coordinate in a lane-candidate-by-row grid. The row part gives the vertical location, while the instance part gives the lane identity. This is why the method should not be interpreted as a generic transformer block inserted into a CNN. The decoder state itself is structured according to the lane geometry.

### H. Row-Local Image Attention

Each lane-row token reads image evidence from the corresponding horizontal row of the feature map. Instead of

Lane-row tokens
↓
Row-local image attention
↓
Same-row lane interaction
↓
Same-lane vertical interaction
↓
Feed-forward update

Fig. 3. Conceptual structure of one decoder block. The final implementation may use standard transformer normalization and feed-forward operations, but the key idea is the separation of row-local, same-row, and vertical interactions.

letting every token attend to the full image, row-local attention restricts the visual search to the row associated with that token. This introduces a useful geometric prior: at a fixed vertical row, the key unknown is the horizontal location of the lane.

Conceptually, for row $j$, the row tokens attend to the horizontal feature sequence:

$$\mathbf{F}_{j,:} = \{\mathbf{f}_{j,k}\}_{k=1}^{M}, \tag{7}$$

where $k$ indexes horizontal positions in the feature map. This enables each lane-row token to collect row-specific visual cues.

### I. Same-Row Lane Interaction

At the same vertical row, multiple lane candidates may compete for nearby horizontal evidence. Same-row interaction allows lane candidates to communicate at a fixed row. This helps reduce duplicate predictions and encourages different candidates to explain different lane structures.

### J. Same-Lane Vertical Interaction

A lane is continuous across rows. Therefore, the row tokens belonging to the same lane candidate exchange vertical context. This helps the model learn curvature, continuity, and occlusion recovery. It also reduces row-wise jitter because each row prediction is informed by neighboring and distant rows along the same lane.

Fig. 3 summarizes the decoder block.

### K. Row-Wise Distributional Localization

Each updated row token predicts a discrete probability distribution over horizontal bins:

$$p_{i,j}(x) = P(x_{i,j} = x \mid \mathbf{z}_{i,j}), \tag{8}$$

where $x$ indexes horizontal bins. The final coordinate can be obtained from the distribution. This representation is more informative than direct scalar regression because it also exposes uncertainty.

For training, the ground-truth coordinate is converted into a soft-bin target. If the true position falls between two neighboring bins, the target mass is linearly distributed between them. This encourages precise localization without forcing a hard one-bin label.

| Bin | $b-2$ | $b-1$ | $b$ | $b+1$ | $b+2$ | $\cdots$ |
|---|---|---|---|---|---|---|
| Target | 0 | 0 | 0.75 | 0.25 | 0 | $\cdots$ |

Fig. 4. Soft-bin supervision for row-wise horizontal localization. If the ground-truth point lies between two bins, both bins receive fractional target mass.

TABLE III
CURRENT HIGH-LEVEL CONFIGURATION OF THE STRONGEST INTERNAL MODEL.

| Component | Setting |
|---|---|
| Backbone | ResNet-34 |
| Input resolution | $1600 \times 640$ |
| FPN channels | 256 |
| Lane candidates | 32 |
| Number of rows | 160 |
| Horizontal bins | 800 |
| Structured decoder layers | 4 |
| EMA | No |
| Test-time augmentation | No |

### L. Lane-Level Prediction Heads

The row tokens of a lane candidate are aggregated into a lane-level representation. This representation predicts:

- existence score: whether the candidate is a lane,
- vertical range: the valid visible region of the lane,
- quality score: the geometric reliability of the prediction.

The row-wise outputs and lane-level outputs are complementary. Row-wise distributions provide geometry, while lane-level scores support filtering and ranking.

### M. High-Level Architecture Configuration

Table III gives the current high-level configuration used for the strongest internal model. These values are listed to clarify the experimental setting, not as the core novelty.

## IV. Training Objective

### A. Assignment

The model predicts a fixed number of lane candidates, while each image contains a variable number of ground-truth lanes. Therefore, predictions must be matched with ground truth during training. The assignment cost combines lane confidence, row-wise localization distance, vertical range consistency, and line-level overlap. Matched predictions are trained as lane instances; unmatched predictions are trained as background or no-lane candidates.

### B. Loss Components

The training objective contains the following conceptual components:

- **Existence loss**: teaches which candidates correspond to real lanes.
- **Point localization loss**: aligns predicted row-wise coordinates with ground truth.
- **Line overlap loss**: encourages the predicted lane as a curve to overlap with the target lane.

- **Range loss**: learns the valid vertical extent of each lane.
- **Smoothness loss**: discourages physically implausible row-to-row jitter.
- **Distributional row loss**: sharpens row-wise horizontal probability distributions around the target position.
- **Auxiliary segmentation and centerline losses**: improve the visual features used by the decoder.

The auxiliary losses are used only to improve training; they do not change the final lane representation.

## V. Inference

At inference time, the model outputs a fixed set of lane candidates. Each candidate has row-wise coordinates and lane-level scores. The final predictions are obtained by:

1) computing lane confidence from existence and quality scores,
2) filtering low-confidence candidates,
3) removing duplicate lanes with lane-level non-maximum suppression,
4) keeping the top-ranked predictions.

The threshold and ranking parameters should be selected on a validation set and then fixed for test evaluation.

## VI. Experiments

### A. Datasets and Metrics

The primary benchmark is CULane. Additional datasets such as TuSimple, LLAMAS, and CurveLanes should be used to evaluate generalization.

The main metrics are precision, recall, and F1 score under the official evaluation protocol. Because the proposed method aims to improve localization, tighter overlap metrics such as F1 at higher IoU thresholds should also be reported. Category-wise results are important for understanding failure modes under normal, crowded, highlight, shadow, no-line, arrow, curve, crossroad, and night scenes.

### B. Validation Protocol

Checkpoint selection, confidence threshold, quality-score weight, and non-maximum suppression parameters must be selected on the validation set. The test set should be used only for final reporting. This is necessary to avoid optimistic test-set tuning.

### C. Implementation Details

Unless otherwise stated, the main model uses a ResNet-34 backbone and is trained from scratch under the same data split as the baseline. The current strongest internal setting uses high-resolution input, a 256-channel feature pyramid, 32 lane candidates, ordered row tokens, and four structured decoder layers. EMA and test-time augmentation are not used in the current preliminary result. This point should be explicitly stated because it separates the architecture result from test-time tricks.

For the final paper, this subsection should report optimizer, learning rate schedule, batch size, augmentation, input resolution, training iterations, and inference settings. These details should be concise but complete enough to make the comparison reproducible.

### D. Comparison with Existing AI Lane Detectors

This section is intended for paper positioning. The numbers in Table IV should be verified from official papers or repositories before final submission, because input resolution, backbone, training schedule, and post-processing may differ across works. The purpose of the table is not to claim final superiority at this stage, but to show where the proposed method sits among established AI-based lane detectors.

Table V compares representation choices rather than only scores. This table is important because the proposed method is not intended to be framed as a pure score-chasing modification. Its claim is representational.

### E. Preliminary Internal Results

Table VII summarizes the current preliminary status. These numbers should be treated as internal results until the final validation-based protocol is fixed.

The preliminary improvement suggests that the structured instance-to-row representation is promising. However, the final claim must be supported by controlled ablations. Otherwise, the improvement could be attributed to input resolution, decoder capacity, distributional supervision, or threshold selection.

### F. CULane Category-Wise Comparison

Total F1 is not sufficient to understand the behavior of a lane detector. CULane contains categories such as normal, crowded, dazzle/highlight, shadow, no-line, arrow, curve, crossroad, and night. Therefore, the paper must not only report the global F1 score; it must show where the model improves and where it still fails. Table VIII follows the detailed CULane reporting style used by strong prior work such as CLRNet and CondLSTR. The Cross column reports the number of false positives because the crossroad split contains images without lane annotations; therefore lower is better for Cross, while higher is better for all other category columns.

The current result is not uniformly better than the strongest public models. This matters for the paper narrative. The proposed decoder is competitive overall and has a low Cross false-positive count in the current run, but it still trails CondLSTR on dazzle, shadow, curve, and night. Therefore, the honest claim is not "we dominate every CULane category." The defensible claim is that the proposed structured row-token representation closes most of the gap to high-performing ResNet-34 methods while avoiding handcrafted line anchors and while providing a clear internal improvement over the unstructured S0 baseline. The category table also defines the next diagnostic target: if future changes improve global F1 but degrade Cross or No-line heavily, they should not be accepted blindly.

TABLE IV

DRAFT COMPARISON WITH REPRESENTATIVE AI-BASED LANE DETECTION MODELS ON CULANE TEST. VALUES SHOULD BE VERIFIED BEFORE FINAL SUBMISSION.

| Method | Venue/Year | Backbone | Paradigm | F1 | Comment |
|---|---|---|---|---|---|
| SCNN [1] | AAAI 2018 | VGG-like | Segmentation / message passing | 71.60 | Early spatial propagation baseline |
| UFLD [2] | ECCV 2020 | ResNet-34 | Row-wise classification | 72.30 | Efficient row-wise formulation |
| LaneATT [3] | CVPR 2021 | ResNet-34 | Anchor-based attention | 76.68 | Strong anchor baseline |
| CondLaneNet [4] | ICCV 2021 | ResNet-34 | Conditional row-wise head | 78.74 | Close instance-first row-wise prior |
| CLRNet [5] | CVPR 2022 | ResNet-34 | Anchor refinement | 79.73 | Strong line-anchor refinement model |
| BSNet | 2023 | ResNet-34 | Spline / curve modeling | 79.89 | Curve-based representation |
| CondLSTR [11] | ICCV 2023 | ResNet-34 | Transformer dynamic kernels | 80.55 | Closest high-performing neighbor |
| Lane2Seq [12] | CVPR 2024 | ViT-Base | Sequence generation | 79.64 | Tokenized output generation |
| **Proposed SRTD** | – | ResNet-34 | Structured row-token decoder | ∼79.9 | No EMA/TTA in current internal result |

TABLE V

ARCHITECTURE-LEVEL COMPARISON WITH SELECTED RELATED METHODS.

| Method | Lane Identity Modeling | Geometry Modeling | Difference from Proposed Method |
|---|---|---|---|
| UFLD | Implicit lane slots | Row-wise grid classification | Efficient row-wise prediction, but no explicit instance-to-row decoder state |
| CondLaneNet | Instance-first proposal | Conditional convolution row-wise shape | Conceptually close, but geometry is generated by conditional heads rather than ordered row tokens |
| CLRNet | Anchor proposals | Iterative line-anchor refinement | Strong refinement model, but relies on predefined line anchors |
| CondLSTR | Transformer lane queries | Dynamic kernels produce row-wise heatmaps/offsets | Strong nearest neighbor; does not explicitly maintain ordered row tokens as decoder states |
| Lane2Seq | Sequence-level tokens | Autoregressive or sequence-style output tokens | Tokenizes outputs, but does not use parallel lane-conditioned row-token geometry |
| **Proposed SRTD** | **Learned instance tokens** | **Ordered row tokens with row-wise distributions** | **Explicitly factorizes lane identity and row geometry inside the decoder** |

TABLE VI

DRAFT PRECISION–RECALL POSITIONING ON CULANE. VALUES SHOULD BE FINALIZED AFTER VALIDATION-BASED THRESHOLD SELECTION.

| Method | Precision | Recall | F1 |
|---|---|---|---|
| LaneATT | 92.14 | 66.47 | 76.68 |
| CondLaneNet | 93.38 | 71.17 | 78.74 |
| CLRNet | 93.49 | 74.57 | 79.73 |
| CondLSTR | – | – | 80.55 |
| **Proposed SRTD** | ∼87.7 | ∼73.4 | ∼79.9 |

TABLE VII

PRELIMINARY CULANE TEST RESULTS WITH RESNET-34. THE FINAL PAPER SHOULD REPORT RESULTS SELECTED USING VALIDATION-SET MODEL SELECTION.

| Model | EMA | TTA | F1 | Note |
|---|---|---|---|---|
| Unstructured S0 baseline | No | No | 75.25 | same-family baseline |
| Structured row-token model | No | No | ∼79.9 | preliminary best |

### G. Cross-Dataset Comparison Plan

CondLSTR- and CLRNet-style papers typically report more than one benchmark or at least include strong CULane split analysis. For a complete submission, the same model family should also be evaluated on TuSimple, LLAMAS, or CurveLanes. Table IX gives the planned reporting format.

### H. Ablation Study Design

The main ablations required for a convincing paper are shown in Table X.

### I. Component Ablation

The final paper should include a compact component ablation similar to the ablation tables used in strong lane detection papers. The goal is to show which part of the method contributes to the final score.

### J. Capacity and Resolution Control

Table XII defines the capacity and resolution controls. These are necessary because a reviewer may argue that the performance gain is due to higher input resolution or a larger decoder rather than the structured representation.

### K. Slot Count and Decoder Depth

The number of lane candidates controls the precision–recall trade-off. Too few slots may reduce recall; too many slots may increase duplicate predictions and false positives. Decoder depth controls the amount of iterative interaction among tokens. The expected behavior is shown in Table XIII.

### L. Efficiency Comparison

Since the proposed representation increases the number of row-level tokens, the final paper should also report parameters, FLOPs, FPS, and memory. This is necessary to avoid presenting only accuracy while hiding computational cost.

TABLE VIII
CULANE CATEGORY-WISE COMPARISON ON THE OFFICIAL TEST SPLIT. PUBLIC METHOD VALUES FOLLOW THE DETAILED CULANE TABLES REPORTED IN CLRNET/CONDLSTR-STYLE PAPERS. THE CROSS COLUMN REPORTS FALSE POSITIVES; LOWER IS BETTER. THE PROPOSED-METHOD ROW IS THE CURRENT INTERNAL BEST AND MUST BE FINALIZED USING VALIDATION-SELECTED CHECKPOINT AND THRESHOLD BEFORE SUBMISSION.

| Method | Backbone | F1 | Normal | Crowd | Dazzle | Shadow | No-line | Arrow | Curve | Cross FP | Night |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SCNN [1] | VGG16 | 71.60 | 90.60 | 69.70 | 58.50 | 66.90 | 43.40 | 84.10 | 64.40 | 1990 | 66.10 |
| UFLD [2] | ResNet-34 | 72.30 | 90.70 | 70.20 | 59.50 | 69.30 | 44.40 | 85.70 | 69.50 | 2037 | 66.70 |
| LaneATT [3] | ResNet-34 | 76.68 | 92.14 | 75.03 | 66.47 | 78.15 | 49.39 | 88.38 | 67.72 | 1330 | 70.72 |
| CondLaneNet [4] | ResNet-34 | 78.74 | 93.38 | 77.14 | 71.17 | 79.93 | 51.85 | 89.89 | 73.88 | 1387 | 73.92 |
| GANet [10] | ResNet-34 | 79.39 | 93.73 | 77.92 | 71.64 | 79.49 | 52.63 | 90.37 | 76.32 | 1368 | 73.67 |
| CLRNet [5] | ResNet-34 | 79.73 | 93.49 | 78.06 | 74.57 | 79.92 | 54.01 | 90.59 | 72.77 | 1216 | 75.02 |
| CondLSTR [11] | ResNet-34 | 80.55 | 94.12 | 79.72 | 77.02 | 82.51 | 53.76 | 90.59 | 76.65 | 1370 | 75.57 |
| **Proposed SRTD** | **ResNet-34** | **79.91** | **93.73** | **78.96** | 73.45 | 80.85 | 51.95 | 89.93 | 74.14 | **1008** | 74.76 |

TABLE IX
PLANNED CROSS-DATASET REPORTING FORMAT.

| Method | CULane F1 | TuSimple Acc. | CurveLanes F1 |
|---|---|---|---|
| Representative prior work | – | – | – |
| Unstructured S0 baseline | – | – | – |
| Proposed SRTD | – | – | – |

### M. Qualitative and Diagnostic Analysis

The paper should include qualitative comparisons and diagnostic plots, not only total F1. The following figures are recommended:

- overall architecture diagram,
- holistic query vs. structured row-token representation,
- instance-to-row token grid,
- decoder interaction diagram,
- row-wise probability heatmaps,
- category-wise gain/loss plot,
- precision–recall or threshold sensitivity curve,
- success and failure visualizations.

Failure cases should be analyzed explicitly. The structured prior may improve continuity in difficult lanes, but it may also hallucinate lanes in no-line or crossroad scenes. This trade-off should be shown rather than hidden.

Fig. 5 summarizes the minimum figure set expected for a complete version of the paper.

## VII. DISCUSSION

The proposed method should be positioned as a representation-level contribution. The main claim is not that high-resolution training or distributional loss alone improves CULane performance. Rather, the key claim is that lane detection benefits from decomposing a lane into global identity and ordered row-wise geometry.

This distinction is important for publication. If the paper is framed as "we used DFL and higher resolution," the contribution will appear incremental. If the paper is framed as "we introduce a structured instance-to-row decoder that changes how lane geometry is represented," the contribution is stronger and more defensible.

## VIII. LIMITATIONS

The method has several limitations:

- It assumes that lane geometry can be represented by one horizontal position per row.
- The strong row-wise lane prior may produce false positives in scenes without visible lane markings.
- One-to-many style training may improve recall but can make confidence calibration harder.
- High row and bin counts increase training cost.
- Final reported results must avoid test-set threshold tuning.

## IX. CONCLUSION

We presented a structured row-token decoder for 2D lane detection. The method decomposes each lane into a global instance token and an ordered sequence of row-level geometry tokens. This representation aligns the decoder state with the natural row-wise structure of lane geometry. Row tokens read row-local visual evidence and produce distributional horizontal predictions, while lane-level heads estimate existence, range, and quality. Preliminary results indicate that this structured representation substantially improves over an unstructured query baseline. Future work should complete controlled ablations, cross-dataset evaluation, backbone scaling, and validation-based threshold selection.

## ACKNOWLEDGMENT

This section is intentionally left as a placeholder.

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[6] L. Liu, X. Chen, S. Zhu, and P. Tan, "LaneFormer: Object-aware row-column transformers for lane detection," in *Proc. AAAI*, 2022.
[7] Y. Feng, L. Xu, Y. Yang, and D. Tao, "BezierLaneNet: A curve-based end-to-end lane detection network," in *Proc. CVPR*, 2022.
[8] X. Li, W. Wang, L. Wu, S. Chen, X. Hu, J. Li, J. Tang, and J. Yang, "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Proc. NeurIPS*, 2020.

TABLE X
CORE ABLATIONS REQUIRED TO ISOLATE THE CONTRIBUTION OF THE PROPOSED REPRESENTATION.

| Variant | Instance Token | Ordered Row Tokens | Distributional Loss | Decoder Depth | Purpose |
|---|---|---|---|---|---|
| Unstructured baseline | Yes | No | No | matched | Tests holistic query representation |
| Instance-only variant | Yes | No | optional | matched | Tests lane identity without row decompositio |
| Structured without distributional loss | Yes | Yes | No | matched | Isolates ordered row-token representation |
| Structured with distributional loss | Yes | Yes | Yes | matched | Tests row-wise distributional localization |
| Shallow structured decoder | Yes | Yes | Yes | 1–2 | Tests insufficient interaction depth |
| Final structured decoder | Yes | Yes | Yes | 4 | Main configuration |
| Deeper structured decoder | Yes | Yes | Yes | 6 | Tests whether extra depth still helps |

| | | |
|---|---|---|
| **Figure A** | Overall architecture | Image $\rightarrow$ backbone/FPN $\rightarrow$ structured decoder $\rightarrow$ lanes |
| **Figure B** | Holistic vs. structured query | Single lane query vs. instance + ordered row tokens |
| **Figure C** | Decoder block | Row-local evidence, same-row interaction, vertical interaction |
| **Figure D** | Probability heatmap | Row index vs. horizontal bin probability |
| **Figure E** | Qualitative results | Baseline vs. proposed success/failure examples |
| **Figure F** | Category-wise delta | Improvement and degradation by CULane category |

Fig. 5. Recommended figure set for the final paper. These placeholders should be replaced by actual diagrams and plots.

TABLE XI
PLANNED COMPONENT ABLATION. THIS TABLE SHOULD BE FILLED WITH VALIDATION-SELECTED TEST RESULTS.

| Variant | Row Tokens | DFL | Aux. Heads | F1 |
|---|---|---|---|---|
| Unstructured baseline | No | No | Yes/No | – |
| Structured, direct regression | Yes | No | Yes | – |
| Structured, distributional | Yes | Yes | Yes | – |
| Structured, no auxiliary heads | Yes | Yes | No | – |
| Final model | Yes | Yes | Yes | – |

TABLE XII
RESOLUTION AND CAPACITY CONTROL EXPERIMENTS.

| Setting | Input | Rows | X bins |
|---|---|---|---|
| Low resolution | 800×288 | 72 | 200 |
| Medium resolution | 1024×384 | 96 | 512 |
| High resolution | 1600×640 | 160 | 800 |


[9] B. Liao, S. Chen, X. Wang, T. Cheng, Q. Zhang, W. Liu, and C. Huang, "MapTR: Structured modeling and learning for online vectorized HD map construction," in *Proc. ICLR*, 2023.

[10] Reference for GANet lane detection. Full bibliographic information should be verified before submission.

[11] Reference for CondLSTR / dynamic-kernel transformer lane detection. Full bibliographic information should be verified before submission.

[12] Reference for Lane2Seq. Full bibliographic information should be verified before submission.


TABLE XIII
RECOMMENDED SLOT-COUNT AND DECODER-DEPTH ABLATIONS.

| Ablation | Expected diagnostic value |
|---|---|
| 20 slots | Cleaner predictions, possible recall loss |
| 32 slots | Balanced setting for CULane-style scenes |
| 64 slots | Higher recall potential, higher duplicate risk |
| 1–2 layers | Tests whether the decoder lacks context |
| 4 layers | Main balance between accuracy and cost |
| 6 layers | Tests saturation and marginal gains |

TABLE XIV
PLANNED EFFICIENCY COMPARISON.

| Method | Params | FLOPs | FPS | Input |
|---|---|---|---|---|
| LaneATT | – | – | – | – |
| CLRNet | – | – | – | – |
| CondLSTR | – | – | – | – |
| Proposed SRTD | – | – | – | 1600×640 |